

TP 4

JavaScript Interactivity (DOM + Events + UI)

Objectives

- Dynamically manipulate DOM elements using JavaScript
- Handle user interactions through events (click, input, submit, keydown)
- Build interactive UI components (slider, FAQ accordion, show/hide sections)
- Implement a Light/Dark theme toggle with icon switching
- Store and restore UI preferences (theme and language) using `localStorage`
- Validate form inputs and display clear error messages
- Implement a basic language switcher (FR/EN) using a translation dictionary and `data-*` attributes
- Create reusable JavaScript features that can be adapted to any website topic

Tools Required

Students will need:

- **VS Code**
- any browser
- DevTools (Console, Elements)
- existing website from previous TP
- recommended project structure:
 - `index.html`
 - `styles.css`
 - `/js/main.js`
 - `/images/`

THEORETICAL INTRODUCTION

JavaScript (JS) is a programming language used to create interactive and dynamic web pages. While HTML defines the structure of a webpage and CSS controls its visual appearance, JavaScript is responsible for the behavior of the website — how it responds to user actions and how content changes without reloading the page. In modern web development, JavaScript runs mainly in the browser on the client side, which means it executes directly on the user's device. Through the DOM (Document Object Model), JavaScript can access and manipulate HTML elements as objects, allowing developers to update text, change styles, add or remove classes, show or hide sections, and create interactive components such as sliders, accordions, tabs, and modal windows.

JavaScript also allows handling events such as clicks, typing, form submissions, scrolling, and keyboard input, making it possible to build responsive and user-friendly interfaces. In addition, JavaScript is commonly used for form validation to ensure that user input is correct before processing, improving both usability and data quality. Another important feature is the ability to store user preferences in the browser using `localStorage`, for example saving the selected theme (dark/light) or language choice so that it remains the same after reloading the page.

Overall, JavaScript is a core technology of front-end development and is essential for creating modern, functional, and engaging web experiences.

1) Connecting JavaScript

At the end of the body:

```
<script src="js/main.js"></script>
```

So the DOM loads before JavaScript runs.

2) Selecting DOM Elements

Most common commands:

```
document.getElementById("id")
document.querySelector(".class")
document.querySelector("#id")
document.querySelectorAll(".item")
```

`querySelectorAll()` returns a list (NodeList). You can loop through it:

```
const items = document.querySelectorAll(".card");

items.forEach((card) => {
  console.log(card);
});
```

3) Changing Text, HTML, Attributes

```
element.textContent = "Hello";
element.innerHTML = "<b>Hello</b>";

img.src = "images/pic.jpg";
input.value = "";
```

4) Working with CSS Classes

```
element.classList.add("active");
element.classList.remove("active");
element.classList.toggle("active");
element.classList.contains("active");
```

5) User Events

Key events:

```
button.addEventListener("click", () => {});  
form.addEventListener("submit", (e) => {});  
input.addEventListener("input", () => {});  
document.addEventListener("keydown", (e) => {});
```

6) Preventing Form Page Reload

Forms reload the page by default.
You must disable it:

```
e.preventDefault();
```

7) localStorage (Saving Theme/Language)

Save:

```
localStorage.setItem("theme", "dark");
```

Load:

```
const theme = localStorage.getItem("theme");
```

Remove:

```
localStorage.removeItem("theme");
```

8) Arrays (for slider and language data)

Example:

```
const slides = [  
  { img: "1.jpg", caption: "Slide 1" },  
  { img: "2.jpg", caption: "Slide 2" }  
];
```

TIPS: To implement a language switcher (FR/EN) for your website, first add a `data-i18n` attribute to every text element that should be translated (for example: titles, menu links, buttons, section headings). Each attribute contains a key such as `data-i18n="nav_home"`. Then, create a translation dictionary in JavaScript (an object) with two languages, for example `translations.en` and `translations.fr`, where each key stores the correct text in that language. After that, write a function `setLanguage(lang)` that selects all elements with `[data-i18n]`, reads their key, and replaces their `textContent` with the translated value from the dictionary. For input placeholders, use a separate attribute like `data-i18n-placeholder` and update the `placeholder` property in the same function. Finally, create a toggle button that switches between languages on click, calls `setLanguage()`, and saves the selected language in `localStorage` so the preference stays after page reload.

TASK : Main Idea (Universal for Any Website)

Each student has a different website topic, but the UI components are universal:

- Slider → “gallery”, “projects”, “services”, “portfolio”, “products”
- FAQ → topic-related questions
- Language → RU/EN (or any)
- Theme → light/dark
- Form → contact / booking / request / order

LEVEL 1 — Basic (Mandatory)

Add minimal interactivity and demonstrate DOM + click event skills.

1) Slider with LOOP (Mandatory)

Requirements:

- minimum **3 slides**
- buttons **Prev / Next**
- when reaching the last slide → go back to the first (loop)
- when going left from the first slide → go to the last (loop)
- display counter: 1 / 3

How to adapt for any website:

- travel website → country photos
- IT portfolio → projects
- medical website → services
- sports website → workouts
- fashion website → collections

2) Night/Day Mode Toggle

Requirements:

- button in header or fixed in top-right corner
- toggles dark or light class on <body>
- icon changes (🌙 / ☀️)

3) FAQ

Requirements:

- minimum 3 questions

- click to open answer
- click again to close

At Level 1, multiple answers can stay open.

Level 1 Result:

The website already feels interactive: slider, theme toggle, and FAQ.

LEVEL 2

Add modern features: localStorage, form validation, show/hide.

4) Save Theme in localStorage

Requirements:

- selected theme is stored
- after page reload, the theme remains the same

5) Form Validation (submit event)

Requirements:

The form must have at least 3 fields:

- Name (min 2 characters)
- Email (must include @ and .)
- Message (min 10 characters)

Errors must be displayed:

- under each field
- in red text
- on successful submit → show green "Sent!" message

The form must not reload the page.

6) Show/Hide Block (Read More)

Requirements:

- button "Show more" / "Hide"
- content block expands and collapses

This can be:

- biography
- service description

- travel details
- product details

Level 2 Result:

The website looks like a real product: theme persists, form works, dynamic content is present.

LEVEL 3

Add features similar to real commercial websites: language switch + UX improvements.

7) Language Toggle RU/EN

Requirements:

- language toggle button: RU | EN
- language is saved in localStorage
- minimum 8 page elements must change text

Recommended elements to translate:

- website title
- navigation menu (Home/About/Services/Contacts)
- “Gallery” section title
- “FAQ” section title
- form submit button “Send”
- input placeholders
- “Show more” button
- FAQ texts

How to implement (logic, not full code):

1. Create translation dictionary:
2. `const translations = {`
3. `ru: { ... },`
4. `en: { ... }`
5. `};`
6. Add a `data-i18n` attribute in HTML:
7. `<h1 data-i18n="title"></h1>`
8. In JS, find all `[data-i18n]` elements and inject the correct language text.

8) FAQ: “Only One Open” Mode

Requirements:

- when opening a new question, the previous one closes automatically

9) Slider: Autoplay + Pause Button

Requirements:

- slides change automatically every 4 seconds
- Pause / Play button
- optional: if user clicks Prev/Next, autoplay resets

10) Close Modal Window with ESC (Optional Bonus)

Add a small modal window:

- “Subscribe”
- “Book now”
- “Contact us”

Level 3 Result:

This becomes a mini interactive web app.
It looks professional and modern.

Checklist

Level 1:

- Slider Next/Prev works
- Loop works correctly
- Theme toggle works
- FAQ opens/closes

Level 2:

- Theme is saved
- Form validation works
- Errors show under inputs
- Show/Hide block works

Level 3:

- RU/EN switch works
- Language is saved
- Minimum 8 elements are translated
- FAQ closes previous item
- Slider autoplay + pause/play works